

Artificial Intelligence and Machine Learning

Detecting AI-Generated Images: A Comparative Study of Machine Learning Approaches

Final Exam

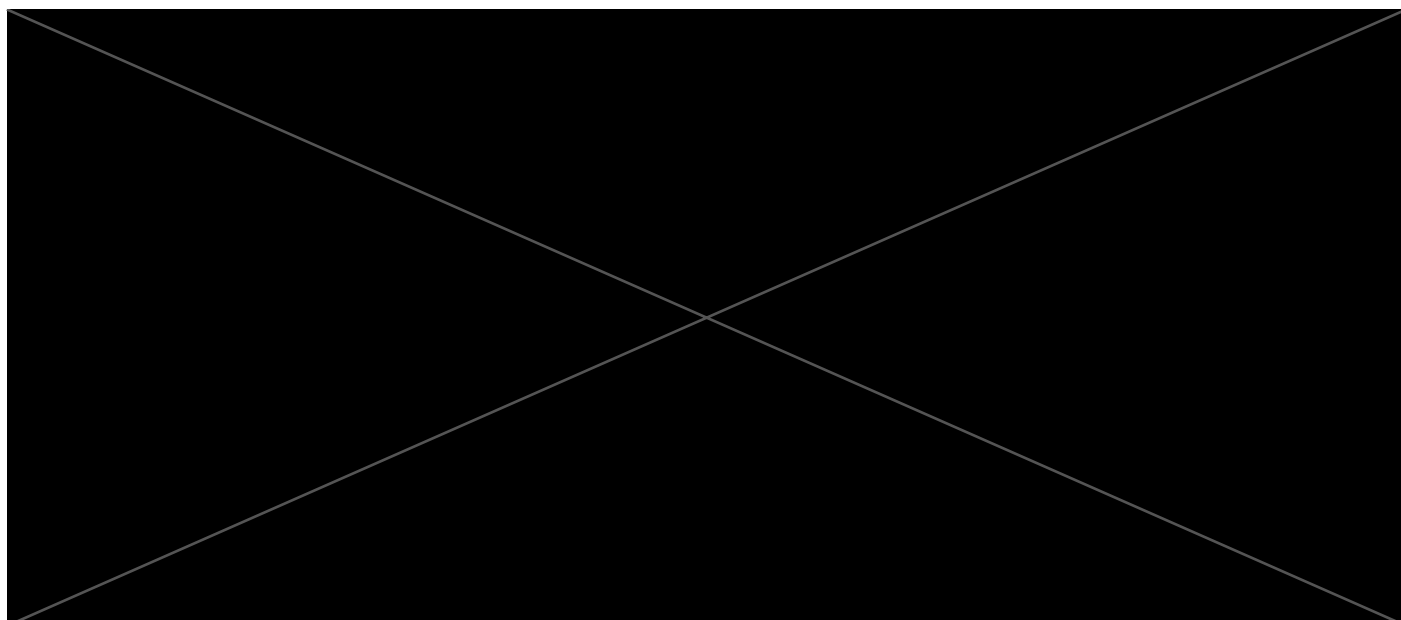


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Introduction

With generative artificial intelligence rapidly advancing, a fundamental transformation of the creation of visual content has followed. Modern text-to-image models such as Stable Diffusion, DALL-E, and MidJourney are now capable of producing highly realistic images, often indistinguishable from real photographs. While these technologies enable new paths of creativity and innovation, they also introduce significant challenges related to misinformation, manipulation, and erosion of trust in visual media. As AI-generated images have become increasingly prevalent across digital platforms, the ability to reliably distinguish between authentic and synthetic content has become a critical task.

Machine learning approaches have shown promise in addressing this challenge. However, effectiveness can depend heavily on model architecture and training data characteristics. This project will investigate the challenges of identifying AI-generated images and focus on the task of binary image classification, where the objective is to determine whether a given image is real or AI-generated. The project is based on a dataset known as MS COCOAI, a large-scale benchmark designed specifically for AI-generated image detection (Roy et al., 2026). Rather than relying on a single modeling approach, this study explores and compares multiple machine learning models and training approaches in order to identify effective methods for AI-generated image detection. Based on this, the guiding research question of this study is:

How effectively can machine learning models distinguish between real and AI-generated images, and how do different modeling approaches influence detection performance?

In the preparation of this paper, ChatGPT was used as a supporting tool for idea development and coding assistance.

Previous Research

Existing research shows that AI-generated image detection (AGID) is a challenging and rapidly evolving task. Many models struggle to generalize to new generators and unseen datasets, making generalization a central issue. For example, *A Sanity Check for AI-generated Image Detection* by Yan et al. (2025) found that state-of-the-art detectors frequently misclassified difficult AI-generated images as real, highlighting the difficulty of the problem.

Much prior work has focused on CNN-based or artifact-driven methods, which often rely on local features and may struggle to capture global semantic inconsistencies. Transformer-based architectures instead use self-attention mechanisms to model relationships across distant image

regions (Vaswani et al., 2017). Recent studies therefore suggest that Vision Transformers (ViT) and CLIP-based models offer improved robustness and generalization, particularly for increasingly realistic and previously unseen AI-generated images (Sar et al., 2025). Additionally, multi-layer aggregation approaches such as MoLD may further improve performance by combining complementary representations across transformer layers rather than relying solely on the final-layer embedding (Park et al., 2025).

Further research is needed to evaluate whether pretrained transformer architectures can improve robustness across multiple AI generators. This project addresses this by systematically comparing linear models, CNNs, pretrained CNNs, Vision Transformers, and MoLD-based architectures on the same dataset, enabling a consistent evaluation of different modeling approaches.

This project builds upon the benchmark results presented by Roy et al. (2026), who are the authors behind the primary dataset used in this paper. This is done for the purpose of comparability, enabling a more direct comparison between results. The baseline study trained a ResNet-50 classifier using image representations in the frequency domain and classified the images into categories as either real or AI-generated. While the baseline results reported in the original benchmark paper provide a useful point of reference for our project, the specific evaluation metric corresponding to the reported performance score of 0.80144 is not explicitly stated (Roy et al., 2026). Given the context of the task and common practice in similar studies (Malviya et al., 2025), we found it reasonable to assume that this score corresponds to F1-score. To ensure a more comprehensive and transparent evaluation, this study reports on multiple metrics, including accuracy, precision, recall, and F1-score. As a result, comparisons with the baseline are primarily used for general performance context rather than exact numerical equivalence.

Data

Dataset Description

The dataset used to answer the research question is known as MS COCOAI and was put together by Roy et al. (2026) for the specific task of distinguishing between authentic and AI-generated images. It contains a large combination of real and artificially generated images and is publicly available via the Huggingface data repository, where it has been obtained from (Roy, 2026). The dataset is built on the widely used MS COCO dataset by Microsoft and other partners for the purpose of developing image recognition tools based on real image photography combined with captions written by human annotators (COCO - Common Objects in Context, 2020). The authors

used images from the original MS COCO dataset and generated a range of AI images using a variety of different image generation models, prompted solely by the original accompanying captions (Roy et al., 2026).

Dataset Composition

The dataset consists of an original 16,000 real images, along with their captions, from the MS COCO dataset. The captions of these images were used by the authors to prompt five different generative models: Stable Diffusion 3 (SD3), Stable Diffusion 2.1 (SD 2.1), SDXL, DALL-E 3 and MidJourney v6 (Roy et al., 2026). This resulted in a total of 96,000 distinct images for classification solely from the MS COCOAI dataset. This approach brings an inherent class imbalance with only $\frac{1}{6}$ of the dataset made up by real images.

In an effort to mitigate the issue of class imbalance, an additional dataset of real images was incorporated solely into the training split (Rawal, 2026), leaving the validation and test sets unchanged to keep the original split. This was done to ensure evaluation would remain consistent with the original dataset distribution and to maintain comparability with our baseline results. Importantly, the original data split is kept as it considers stratification, where images with the same caption are kept together in the same split to minimize potential data leakage. The inclusion of additional real images improves class balance during training, with the purpose of enabling the models to better learn features associated with authentic images. However, it also introduces a potential distribution shift between training and evaluation data. This approach results in the final dataset being divided into:

- Training Set: 52,666 images (7,000 real + 10,666 additional real and 35,000 AI-generated)
- Validation Set: 9,000 images (1,500 real and 7,500 AI-generated)
- Test Set: 45,000 images (7,500 real and 37,500 AI-generated)

Data Structure and Labels

The dataset is organized into a simple structured format where each sample consists of multiple fields. Each data point includes: *id* (unique identifier), *image*, *caption* (MS COCO prompt), *label_1* (0 = real, 1 = AI-generated), and *label_2* (generative model).

While the original dataset provides a standardized structure, the additional real images introduced during training were integrated to match this format, ensuring consistency in label encoding and data handling across all samples.

Importantly, a characteristic of the dataset is that all images are stored at their native resolution without prior normalization or resizing (Roy et al., 2026). This means that any preprocessing required for model training must be performed as part of the data pipeline and will be covered in the next section.

Strengths and Limitations

A key strength of the dataset is its semantic alignment, where real and AI-generated images are derived from identical captions (Roy et al., 2026). This design enables controlled comparison between authentic and synthetic images, which we hope will allow the model to focus on actual generation artifacts rather than differences in image content. Additionally, the dataset includes a healthy variety of images generated by multiple state-of-the-art models (Roy et al., 2026). This exposes the model to healthy diversity in relation to generative style and improves its ability to learn more generalizable patterns. This is further supported by the large scale of the dataset.

However, several limitations must also be considered. As previously mentioned, there is a large imbalance in the dataset, with a significantly larger proportion of AI-generated images compared to real images. Although this imbalance was partially mitigated by the incorporation of additional real images during training, the evaluation metrics remain greatly imbalanced, which could influence performance metrics. This issue was further mitigated via the introduction of weighted sampling. Moreover, the introduction of additional images creates a distribution shift between training and evaluation data, which may have the potential to affect the results (Bialek, 2023).

Finally, a major limitation that was faced during this project, however overcome, was an initial mislabeling issue with the test data. During the initial augmentation of the test dataset, performed by the original authors, a mislabeled JSON file caused the test split to be incorrectly labeled. We discovered this issue during our initial model evaluation due to a large unexplainable discrepancy between the validation and test scores. We flagged this issue for the authors of the dataset and it was subsequently corrected.

Modeling

Task, Target, and Modeling Objective

The modeling task is framed as a binary image classification problem, where the objective is to distinguish real images from AI-generated images. We use Label_A as the primary target variable, with 0 representing real images and 1 representing AI-generated images, while Label_B is retained for later generator-level analysis (Roy, 2026).

Modeling Strategy and Model Selection

Our modeling strategy was to compare six model pipelines of increasing complexity, from traditional machine learning (logistic regression) to pretrained transformer-based approaches. This also allowed us to compare different modeling paradigms for the same task, including traditional classification, convolutional neural networks and vision transformer-based models. This progression made it possible to observe how performance changed as the models gained access to spatial feature learning, pretrained visual representations and all-layer feature aggregation. This approach also helped us determine if more advanced and complex methods were justified to proceed with.

While all six models are included as comparison points, we focus in depth on the Custom CNN and CLIP-MoLD in this section. The Custom CNN was selected because it is trained from scratch and represents a transparent deep learning approach based on convolutional feature learning (Krizhevsky et al., 2012). CLIP-MoLD was selected because it is our strongest performing pipeline and combines pretrained CLIP representations with an all-layer MoLD head (Park et al., 2025). The remaining models support the overall comparison but are not analysed in the same depth.

Model Configuration Overview

The table below summarizes the key configuration choices across the implemented model pipelines. The purpose is not to include every implementation detail, but to show the modeling choices that affect comparability, model capacity, and training behaviour. All neural models were trained with seed 42. For all neural models, weight decay was set to 1e-4 and a weighted sampler was applied. The decision threshold was set to 0.35.

Model	Pretrained	Freeze backbone	Pretrained model	Epochs	LR	Batch size	Optimizer	Dropout
Logistic Regression	No	N/A	N/A	N/A	N/A	N/A	LBFGS	N/A
Custom CNN	No	N/A	N/A	15	1e-3	32	Adam	00.30
ResNet18	Yes	FALSE	ImageNet ResNet18	15	1e-4	32	Adam	N/A
ViT Base	Yes	FALSE	vit_base_patch16_224	15	2e-5	16	AdamW	00.30
ViT-MoLD	Yes	FALSE	vit_base_patch16_224	15	2e-5	16	AdamW	00.30
CLIP-MoLD	Yes	TRUE	openai/clip-vit-large-patch14	15	1e-4	16	AdamW	00.30

Model-specific architectural parameters, such as convolutional channel sizes in the Custom CNN and projection dimensions or layer aggregation mechanisms in the MoLD models, are discussed in the respective model deep dives rather than included in the overview table.

Data Preprocessing

This section focuses on the neural network based models preprocessing methods. Across the neural models, preprocessing was treated as part of the modeling design rather than only as a technical step. Since our exploratory data inspection indicated systematic differences in image size, aspect ratio and formatting between real and AI-generated images, the input images were standardized before training. For the neural-network-based models (except the CLIP-MoLD model), the preprocessing pipeline consisted of:

- Converting all images to three-channel RGB format
- Resizing images to 256 pixels
- Center-cropping images to standardized 224×224 inputs
- Applying training augmentation through random horizontal flips, random rotation, and color jitter
- Converting images to PyTorch tensors
- Normalizing tensors using ImageNet mean and standard deviation values

CLIP-MoLD follows the same overall standardization logic, but differs slightly, because it uses CLIP-specific preprocessing: images are resized to 224 pixels, center-cropped to 224×224, and normalized using CLIP mean and standard deviation values. This was necessary because the CLIP backbone expects inputs in the format used during pretraining (Radford et al., 2021).

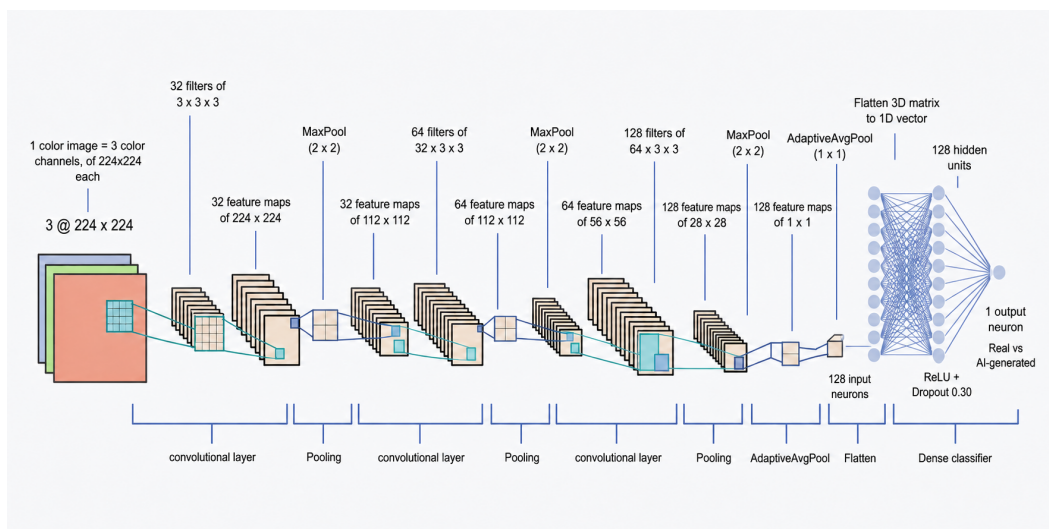
This preprocessing reduces the risk of models learning shortcuts based on image dimensions, aspect ratio, and/or formatting, rather than visual indicators of AI generation. The training augmentations introduce limited visual variation through flips, rotations, and color jitter, but they do not explicitly simulate real-world post-processing noise, such as Gaussian noise, JPEG compression, blur, or social-media recompression. Therefore, preprocessing cannot fully eliminate dataset biases, such as generator-specific artifacts, compression differences, or content imbalance.

Deep Dive: Custom CNN

The Custom CNN was selected for deeper analysis because it represents a transparent neural-network model trained from scratch (Krizhevsky et al., 2012). Unlike the pretrained models, it

does not rely on external visual representations, making it useful for examining how much can be learned directly from the dataset through convolutional feature learning.

The model consists of three convolutional blocks with 32, 64, and 128 output channels. Three blocks were chosen to give the model the ability to gradually learn more complex patterns in the images, and the number of output channels was increased, such that the model could learn increasingly complex patterns deeper in the network. Each block applies a convolutional layer, batch normalization, ReLU activation, and max pooling. Max pooling is done to reduce the feature map size, but to keep the strongest signals. Batch normalization was done to stabilize the training. The ReLU activation was implemented to add non-linearity. The feature extractor is followed by adaptive average pooling and a dense binary classifier with 128 hidden units, ReLU activation, dropout of 0.30 and a final single-output layer. The hidden layer gives the classifier capacity to combine the extracted visual features in a non-linear way before producing the final real-vs-AI prediction. Dropout of 0.30 was chosen to reduce overfitting by randomly shutting off 30% of nodes during training. This reduces the chance of overreliance on certain nodes, thereby reducing overfitting. The model was trained for 15 epochs using Adam with a learning rate of $1e-3$ and weight decay of $1e-4$. Binary classification was trained with BCEWithLogitsLoss, and a weighted sampler was used to address class imbalance during training.

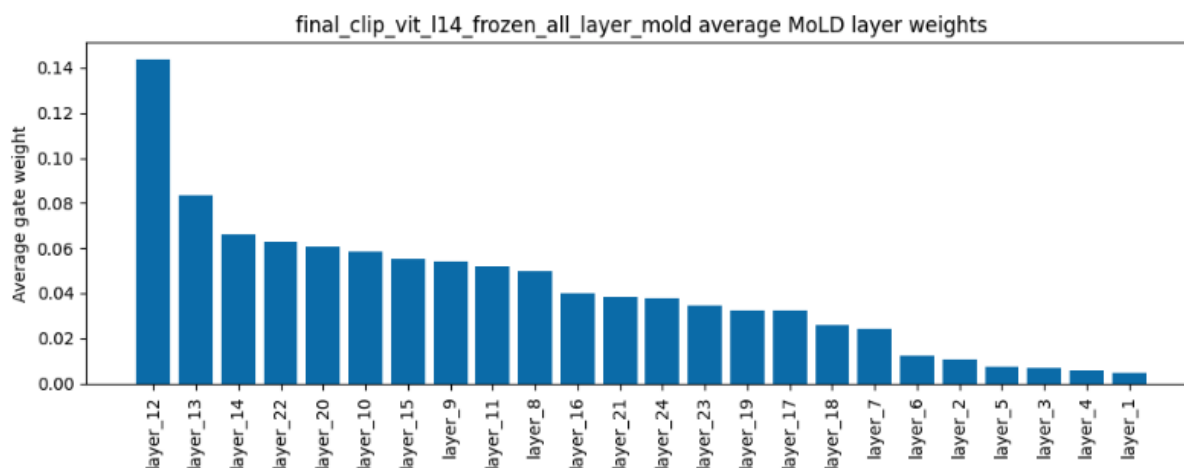


Deep Dive: CLIP-MoLD

CLIP-MoLD was selected for deeper analysis because it represents the strongest advanced pipeline in the project and provides a clear contrast to the Custom CNN. Whereas the Custom CNN learns visual features from scratch, CLIP-MoLD uses a frozen pretrained CLIP vision encoder as a feature extractor (Radford et al., 2021).

This means that the CLIP backbone weights are not updated during training. Instead, the model builds on visual representations learned during CLIP pretraining, while only the task-specific projection layers, layer-gating mechanism, and classifier head are trained for the real-vs-AI classification task.

The MoLD component uses feature representations from all 24 CLIP vision transformer layers rather than relying only on the final layer. Each layer provides a different numerical representation of the image, potentially capturing different levels of visual information. Earlier layers may contain lower-level information such as colours, textures and edges, while later layers may capture more abstract image structure (Radford et al., 2021). To combine these representations, each layer output is projected into a shared 512-dimensional space. A learned softmax gate then assigns weights to the different layers, allowing the model to learn which parts of the CLIP vision encoder are most useful for detecting AI-generated images. The graph below visualizes the average gate weight for every layer, demonstrating which layers are most relevant for the task.



The weighted layer representations are combined and passed to a binary classifier with dropout of 0.30 and a final single-output layer. The output is a logit, which is converted into a probability for the AI-generated class. This design is relevant because AI-image detection may depend on several levels of visual information, from low-level artifacts to higher-level inconsistencies in image structure (Park et al., 2025). However, CLIP-MoLD should be interpreted as a full pipeline rather than isolated evidence that MoLD alone explains the performance. The model also differs from the other pipelines in backbone size, CLIP pretraining, frozen-backbone setup, and CLIP-specific preprocessing.

Modeling Limitations

Pipeline Comparison: First, the comparison should be interpreted as a comparison of full pipelines rather than a perfectly controlled architecture experiment. The models differ not only in architecture, but also in pretraining, backbone size, number of trainable parameters, preprocessing, and training setup. This means that stronger performance cannot always be attributed to one isolated modeling component. For example, CLIP-MoLD should be understood as the strongest full pipeline rather than as proof that the MoLD mechanism alone explains the performance.

Generator-Specific Learning: Second, the models may learn generator-specific artifacts rather than general indicators of AI-generated imagery. Since the dataset contains images from a limited set of known generators, the models may partly learn patterns associated with these specific systems. Strong performance on the held-out test split, therefore, does not necessarily imply robustness to unseen generators or future image generation models.

Preprocessing Tradeoff: Third, preprocessing introduces a trade-off. Standardizing the images through resizing and center cropping reduces the risk that models learn shortcuts based on image size, format, or aspect ratio. However, this preprocessing may also remove visual context or fine-grained details that could be relevant for detecting AI-generated images. Preprocessing, therefore, improves comparability across models, but it is not a neutral modeling choice.

No Uncertainty Handling: Finally, the modeling setup does not include explicit uncertainty handling. Each model produces a binary prediction using a fixed threshold, but there is no abstention mechanism for borderline cases. In practical applications, uncertain predictions may require manual review rather than being treated as confident real-vs-AI classifications. Interpretability is also a relevant limitation, but this is discussed further in the discussion section due to its broader implications for trust and deployment.

Results

The results show a clear performance progression from simpler baselines to pretrained transformer-based models. Since the test set is imbalanced toward AI-generated images, the most useful overall comparison is based on macro precision, macro recall, macro F1, and accuracy, rather than accuracy alone. The strongest overall model is CLIP-MoLD, followed by All-Layer ViT-MoLD as the strongest non-CLIP model.

Overall Results by Model

Model	Precision	Recall	F1	Accuracy
Logistic Regression	0,530	0,521	0,521	0,772
Custom CNN	0,761	0,805	0,779	0,867
ResNet18	0,867	0,944	0,899	0,938
Vision Transformer	0,919	0,971	0,942	0,966
All-Layer ViT-MoLD	0,927	0,975	0,949	0,970
CLIP-MoLD	0,953	0,989	0,970	0,983

CLIP-MoLD achieved the best overall performance, with the highest macro precision, macro recall, macro F1, and accuracy. The results also show that the pretrained models outperformed the simpler baselines substantially. Logistic regression performed weakest, while the custom CNN improved the results by preserving spatial image structure. ResNet18 improved further through pretraining, and the transformer-based models produced the strongest results overall. All-Layer ViT-MoLD improved slightly over the standard ViT, suggesting that combining information from multiple transformer layers provided some additional value.

Per-Generator AI Results by Model

The following table compares each model's AI recall by generator, meaning the percentage of AI-generated images from each generator that the model correctly classified as AI-generated. A higher percentage means the model was better at detecting images from that generator, while the opposite percentage (which indicates how many pictures were misclassified) is implied. For example, if a model has 92.4% recall for Midjourney, then 7.6% of Midjourney images were incorrectly classified as real. This helps show whether the model performs consistently across AI generators or whether certain generators were harder to detect.

Model	SD21	SDXL	SD3	DALLE3	Midjourney
Logistic Regression	87.6%	93.6%	90.9%	87.5%	89.5%
Custom CNN	86.4%	88.1%	89.4%	94.9%	89.8%
ResNet18	87.3%	96.6%	94.7%	98.6%	89.9%
Vision Transformer	92.3%	98.7%	97.8%	99.4%	93.4%
All-Layer ViT-MoLD	92.6%	99.3%	97.4%	99.6%	94.7%
CLIP-MoLD	98.4%	99.8%	99.6%	99.6%	92.4%

Discussion

Evaluation Metrics

In the context of AI-generated image detection, we consider precision and recall as equally important evaluation criteria. A model with high precision but low recall may classify images very conservatively, correctly identifying only a limited subset of AI-generated or authentic images while missing many relevant cases. Conversely, a model with high recall but low precision may identify most manipulated images, but at the cost of producing a large number of false classifications. In practical applications, both types of errors can have significant consequences. Incorrectly classifying authentic images as AI-generated may reduce trust and credibility, while failing to detect synthetic content may contribute to misinformation and manipulation.

For this reason, the project does not rely on a single primary evaluation metric. Instead, multiple complementary metrics are reported, including precision, recall, F1-score, and macro F1, in order to provide a more balanced and comprehensive assessment of model performance. This approach is particularly important given the imbalanced nature of the dataset and the broader objective of evaluating both robustness and generalization across different AI-generated image types. Furthermore, the overall ranking and relative performance differences between the models remain largely consistent across the reported metrics. No major shifts in model performance are observed depending on whether evaluation is based on precision, recall, or F1-score, suggesting that the conclusions of the project are robust across multiple evaluation perspectives rather than being dependent on a single selected metric.

Discussion of Results

The results demonstrate a clear and consistent performance progression across the evaluated model architectures. Logistic Regression achieved the weakest performance across all evaluation metrics, indicating that flattened pixel representations are insufficient for capturing the complex visual patterns required for reliable AI-generated image detection. The Custom CNN improved performance considerably, showing that convolutional architectures are capable of learning meaningful spatial image features directly from the data (Krizhevsky et al., 2012). However, its performance remained substantially below the pretrained architectures.

ResNet18 achieved a significant improvement over the Custom CNN, demonstrating the benefit of pretrained convolutional representations learned from large-scale image datasets. The model achieved strong results across precision, recall, F1-score, and accuracy, suggesting that

pretrained representations provide substantially more robust visual feature extraction than training a CNN from scratch.

However, the transformer-based architectures consistently achieved the strongest overall performance. Vision Transformer (ViT), All-Layer ViT-MoLD, and CLIP-MoLD all outperformed the CNN-based models across every reported metric. One possible explanation for this difference lies in the underlying architectural properties of transformers. CNNs primarily rely on local convolutional filters and progressively expanding receptive fields, meaning that feature extraction is focused on localized spatial patterns. Transformer-based models instead utilize self-attention mechanisms, allowing relationships between distant image regions to be modeled simultaneously (Vaswani et al., 2017).

The stronger performance of the transformer-based models, therefore, supports existing research suggesting that transformers learn richer and more generalized semantic representations compared to traditional CNN-based approaches (Park et al., 2025). Recent studies on zero-shot deepfake detection further argue that transformer architectures may generalize better to increasingly realistic and previously unseen AI-generated content, which aligns closely with the findings of this project (Sar et al., 2025).

While the MoLD-based architectures achieved the strongest overall results, the improvements over the standard ViT baseline were relatively incremental rather than transformative. This suggests that the primary performance gain stems from the transformer-based representation learning itself, while the MoLD aggregation mechanism mainly contributes additional robustness and refinement rather than fundamentally different capabilities (Park et al., 2025).

Comparison to Related Work

Direct comparison with related studies should be made with caution, as AI-generated image detection results depend heavily on dataset choice, image generators, preprocessing, train-test splits, class balance, metrics, and training strategy. Therefore, numerical scores are not always directly comparable across studies. Instead, related work is most useful for identifying broader trends, such as the strong performance of pretrained transformer- and CLIP-based models compared with simpler CNN or traditional machine learning baselines. This trend is also reflected in our results. However, as our future work section proposes, unseen-generator and cross-dataset testing would be required before making broader comparisons.

Comparison To Baseline

Compared to the baseline study by Roy et al. (2026), the transformer-based architectures evaluated in this project achieved substantially stronger overall performance across most reported metrics. While the original benchmark reported a performance score of 0.80144 using a frequency-domain ResNet-50 classifier, the ViT and CLIP-MoLD architectures in this project achieved notably higher Macro F1 scores. In particular, the CLIP-MoLD model demonstrated very strong balanced classification performance and robustness across the benchmark dataset. However, direct numerical comparison should still be interpreted cautiously due to differences in evaluation setup, preprocessing strategies, class balancing, and reported metrics between the studies.

Ethics

Although our AI image recognition model is designed for a narrow binary classification task (predicting whether an image is real or AI-generated), it still raises ethical concerns related to fairness, privacy, misuse, and transparency. The model should therefore be understood as a decision-support and research tool, not as a definitive judge of image authenticity.

One concern is fairness and potential harm from misclassification. Although the model does not directly classify protected attributes. If the training dataset is not representative, the model may become more likely to classify certain people, objects, or settings as AI-generated than others. This creates a risk of unequal harm, especially if real images from underrepresented contexts are falsely labeled as AI-generated. This reflects the broader issue of upstream bias creating downstream bias: if the real or AI-generated images used for training already contain biased patterns, the model can inherit these biases before training even begins. Resnik (2025) makes a similar argument in the context of LLMs, where models trained on human-generated data can absorb harmful patterns that are difficult to remove later.

A second concern is misuse. A real/AI detector could be useful for research, education, or content analysis, but it could also be misused as an automated enforcement tool. For example, it could be used to accuse students of cheating, remove social media posts, reject images, or question journalistic evidence without human review. This is problematic because the model is probabilistic and imperfect. A high-confidence prediction is not proof; it only shows that an image resembles patterns learned during training. For this reason, the EU AI Act emphasizes that human oversight should help prevent or minimize risks to health, safety, and fundamental rights (European Parliament and Council of the European Union, 2024). In our case, the model should support human judgment rather than replace it.

Finally, the black-box nature of deep learning creates transparency concerns. Users and developers may want to know why a specific image was classified as real or AI-generated, especially when the assessment is wrong. This is the purpose of explainable AI (XAI), which aims to make model decisions more understandable. Kashefi et al. (2023) explain that XAI can help identify biases, errors, and limitations in deep neural networks, and they distinguish between global methods, which explain broader model behavior, and local methods, which explain individual predictions. In this project, we address transparency in a limited way through confidence scores, threshold selection, confusion matrices, class-level metrics, AI-generator-level metrics, and gate-weight analysis (see *Appendix for detailed breakdown of model results*). However, these are not complete explanations. Further work could apply methods such as saliency maps, Grad-CAM, Integrated Gradients, SHAP, attention rollout, or attention flow, while recognizing that attention-based explanations should be treated cautiously because attention is only one part of a complex model pipeline (Kashefi et al., 2023; Bhagya et al., 2023).

Future Research

Future research should primarily focus on improving real-world generalization. Although the models are evaluated on a held-out test split, this does not fully demonstrate whether they can detect images from unseen generators, future model versions, or different datasets. Future work should therefore include leave-one-generator-out testing, cross-dataset validation and evaluation on entirely unseen generators. Robustness should also be tested under common real-world transformations such as compression, screenshots, cropping, resizing, and social media re-uploading.

Another important direction is to move beyond strict binary classification. The current task assumes that images are either real or AI-generated, but many real-world images are likely to be hybrid cases. Examples include real photographs edited with AI tools, AI-generated images manually modified after generation, or real images containing AI-generated elements. Future datasets and models should therefore consider more nuanced categories that better reflect how images are created and edited in practice.

Our results indicate that model performance varies across different AI generators. Future research should examine why some generator outputs are harder to detect than others, for example, by analysing generator-specific artifacts, image quality, and similarity to real photographs. In addition to our generator-level analysis, future work could examine whether certain image types are harder to detect than others. For example, portraits, landscapes, product images, text-heavy images, or highly stylized images may produce different error patterns. This could provide valuable insight into where AI-image detectors are most reliable and where they

are most likely to fail. Finally, future work could strengthen explainability and uncertainty handling. Explainable AI methods could help identify whether models rely on meaningful visual cues or dataset-specific shortcuts.

Conclusion

This study investigated how effectively machine learning models can distinguish between real and AI-generated images, and how different modeling approaches influence detection performance. The results show that models can achieve strong performance on this task, particularly when using pretrained representations.

A clear progression was observed across models, with simpler approaches such as logistic regression and the custom CNN performing weakest, while pretrained models, especially transformer-based architectures, consistently achieved the best results. This suggests that advanced feature representations and global context modeling play a crucial role in effective AI-generated image detection.

However, the results are influenced by dataset characteristics, including class imbalance and the limited set of generators. Consequently, strong performance on the benchmark dataset does not necessarily imply robustness to unseen models or real-world conditions.

In conclusion, machine learning models can effectively distinguish between real and AI-generated images in controlled settings. However, their performance remains dependent on both model architecture and dataset properties, highlighting the need for further research into generalization and real-world robustness.

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Appendix

Appendix A: Model-by-Model Results

Appendix A1: Logistic Regression

Field	Value
Notebook	02_logistic_regression_baseline_model_STANDARDIZED.ipynb
Final model name	final_logistic_regression_64_stronger_reg
Architecture	Scikit-learn logistic regression on flattened pixel vectors

Final Parameters

Parameter	Value
Image size	64 x 64
Input features	12,288 flattened pixel features
C	0.1
Class weight	None
Threshold	0.35
Scaler	StandardScaler
Training	Fit once on full training split

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.220	0.144	0.174	7,500
AI-generated	0.840	0.898	0.868	37,500
Accuracy			0.772	45,000
Macro average	0.530	0.521	0.521	45,000
Weighted average			0.752	45,000

Summary Statistics

Metric	Value
Accuracy	0.772
Real precision	0.220
Real recall	0.144
Real F1	0.174
AI precision	0.840
AI recall	0.898
AI F1	0.868

Macro precision	0.530
Macro recall	0.521
Macro F1	0.521
Weighted F1	0.752
ROC AUC	0.554

Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
Real	1,081	6,419
AI-generated	3,826	33,674

In real terms, this model correctly classified 1,081 of 7,500 real images as Real and incorrectly flagged 6,419 real images as AI-generated. It correctly caught 33,674 of 37,500 AI-generated images and missed 3,826 AI-generated images by labeling them Real. Real precision of 0.220 means that when the model predicted Real, it was correct 22.0% of the time. AI precision of 0.840 means that when the model predicted AI-generated, it was correct 84.0% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
SDXL	7,500	7,017	483	0.936	0.678	0.708

SD3	7,500	6,815	685	0.909	0.722	0.784
Midjourney	7,500	6,711	789	0.895	0.668	0.702
SD21	7,500	6,572	928	0.876	0.666	0.710
DALLE3	7,500	6,559	941	0.875	0.682	0.735

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
Real	7,500	0.640	0.676	0.00934	0.99996
AI-generated	37,500	0.683	0.726	4.4e-05	0.999994

Appendix A2: Custom CNN

Field	Value
Notebook	03_cnn_custom_model_STANDARDIZED.ipynb
Final model name	final_custom_cnn_baseline
Architecture	Custom convolutional neural network trained from scratch

Final Parameters

Parameter	Value
Max epochs	15

Learning rate	1e-3
Weight decay	1e-4
Threshold	0.35
Batch size	32
Weighted sampler	True
Conv channels	(32, 64, 128)
Dense units	128
Dropout	0.30
Optimizer	Adam
Loss	BCEWithLogitsLoss

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.581	0.714	0.641	7,500
AI-generated	0.940	0.897	0.918	37,500
Accuracy			0.867	45,000
Macro average	0.761	0.805	0.779	45,000

Weighted average			0.872	45,000
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Summary Statistics

Metric	Value
Accuracy	0.867
Real precision	0.581
Real recall	0.714
Real F1	0.641
AI precision	0.940
AI recall	0.897
AI F1	0.918
Macro precision	0.761
Macro recall	0.805
Macro F1	0.779
Weighted F1	0.872
ROC AUC	0.918

Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
Real	5,352	2,148
AI-generated	3,858	33,642

In real terms, this model correctly classified 5,352 of 7,500 real images as Real and incorrectly flagged 2,148 real images as AI-generated. It correctly caught 33,642 of 37,500 AI-generated images and missed 3,858 AI-generated images by labeling them Real. Real precision of 0.581 means that when the model predicted Real, it was correct 58.1% of the time. AI precision of 0.940 means that when the model predicted AI-generated, it was correct 94.0% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
DALLE3	7,500	7,119	381	0.949	0.908	0.997
Midjourney	7,500	6,734	766	0.898	0.846	0.986
SD3	7,500	6,702	798	0.894	0.811	0.947
SDXL	7,500	6,609	891	0.881	0.841	0.985
SD21	7,500	6,478	1,022	0.864	0.798	0.955

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
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Real	7,500	0.254	0.103	9e-06	0.998519
AI-generated	37,500	0.841	0.980	1.5e-05	1

Appendix A3: Finetuned ResNet18 CNN

Field	Value
Notebook	03_cnn_resnet18_model_finetuned_STANDARDIZED.ipynb
Final model name	final_resnet18_unfrozen_linear
Architecture	ImageNet-pretrained ResNet18 fine-tuned with a linear binary head

Final Parameters

Parameter	Value
Max epochs	15
Learning rate	1e-4
Weight decay	1e-4
Threshold	0.35
Batch size	32
Weighted sampler	True
Freeze backbone	False

Head type	Linear
Optimizer	Adam
Loss	BCEWithLogitsLoss

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.744	0.953	0.836	7,500
AI-generated	0.990	0.934	0.961	37,500
Accuracy			0.938	45,000
Macro average	0.867	0.944	0.899	45,000
Weighted average			0.941	45,000

Summary Statistics

Metric	Value
Accuracy	0.938
Real precision	0.744
Real recall	0.953

Real F1	0.836
AI precision	0.990
AI recall	0.934
AI F1	0.961
Macro precision	0.867
Macro recall	0.944
Macro F1	0.899
Weighted F1	0.941
ROC AUC	0.987

Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
Real	7,151	349
AI-generated	2,460	35,040

In real terms, this model correctly classified 7,151 of 7,500 real images as Real and incorrectly flagged 349 real images as AI-generated. It correctly caught 35,040 of 37,500 AI-generated images and missed 2,460 AI-generated images by labeling them Real. Real precision of 0.744 means that when the model predicted Real, it was correct 74.4% of the time. AI precision of 0.990 means that when the model predicted AI-generated, it was correct 99.0% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
DALLE3	7,500	7,395	105	0.986	0.975	1.000
SDXL	7,500	7,248	252	0.966	0.948	1.000
SD3	7,500	7,103	397	0.947	0.925	1.000
Midjourney	7,500	6,744	756	0.899	0.870	0.999
SD21	7,500	6,550	950	0.873	0.838	0.998

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
Real	7,500	0.045	0.000	6.15363e-19	0.999973
AI-generated	37,500	0.911	1.000	4.35824e-09	1

Appendix A4: Finetuned base_patch16_224 Vision Transformer (ViT)

Field	Value
Notebook	04_vit_base_patch16_224_model_finetuned_STANDARDIZED.ipynb
Final model name	final_vit_base_patch16_224_unfrozen_mlp
Architecture	Pretrained ViT-Base fine-tuned with an MLP binary head

Final Parameters

Parameter	Value
Max epochs	15
Learning rate	2e-5
Weight decay	1e-4
Threshold	0.35
Batch size	16
Weighted sampler	True
Freeze backbone	False
Head type	MLP
Optimizer	AdamW
Loss	BCEWithLogitsLoss

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.841	0.980	0.905	7,500
AI-generated	0.996	0.963	0.979	37,500

Accuracy			0.966	45,000
Macro average	0.919	0.971	0.942	45,000
Weighted average			0.967	45,000

Summary Statistics

Metric	Value
Accuracy	0.966
Real precision	0.841
Real recall	0.980
Real F1	0.905
AI precision	0.996
AI recall	0.963
AI F1	0.979
Macro precision	0.919
Macro recall	0.971
Macro F1	0.942
Weighted F1	0.967

ROC AUC	0.997
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Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
Real	7,347	153
AI-generated	1,385	36,115

In real terms, this model correctly classified 7,347 of 7,500 real images as Real and incorrectly flagged 153 real images as AI-generated. It correctly caught 36,115 of 37,500 AI-generated images and missed 1,385 AI-generated images by labeling them Real. Real precision of 0.841 means that when the model predicted Real, it was correct 84.1% of the time. AI precision of 0.996 means that when the model predicted AI-generated, it was correct 99.6% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
DALLE3	7,500	7,454	46	0.994	0.993	1.000
SDXL	7,500	7,401	99	0.987	0.984	1.000
SD3	7,500	7,338	162	0.978	0.973	1.000
Midjourney	7,500	7,002	498	0.934	0.926	1.000
SD21	7,500	6,920	580	0.923	0.912	1.000

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
Real	7,500	0.019	0.000	1.41778e-07	0.999995
AI-generated	37,500	0.957	1.000	2.53711e-07	1

Appendix A5: Finetuned All-Layer base_patch16_224 Vision Transformer (ViT) MoLD

Field	Value
Notebook	05_vit_all_layer_mold_model_STANDARDIZED.ipynb
Final model name	final_vit_all_layer_mold_unfrozen_low_lr
Architecture	Pretrained ViT-Base with all-layer gated MoLD head

Final Parameters

Parameter	Value
Max epochs	15
Learning rate	2e-5
Weight decay	1e-4
Threshold	0.35

Batch size	16
Weighted sampler	True
Freeze backbone	False
Projection dim	512
Dropout	0.30
Layers used	All 12 ViT transformer blocks
Combination method	Learned softmax gate
Optimizer	AdamW
Loss	BCEWithLogitsLoss

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.857	0.983	0.916	7,500
AI-generated	0.997	0.967	0.982	37,500
Accuracy			0.970	45,000
Macro average	0.927	0.975	0.949	45,000
Weighted average			0.971	45,000

Summary Statistics

Metric	Value
Accuracy	0.970
Real precision	0.857
Real recall	0.983
Real F1	0.916
AI precision	0.997
AI recall	0.967
AI F1	0.982
Macro precision	0.927
Macro recall	0.975
Macro F1	0.949
Weighted F1	0.971
ROC AUC	0.998

Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
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Real	7,374	126
AI-generated	1,231	36,269

In real terms, this model correctly classified 7,374 of 7,500 real images as Real and incorrectly flagged 126 real images as AI-generated. It correctly caught 36,269 of 37,500 AI-generated images and missed 1,231 AI-generated images by labeling them Real. Real precision of 0.857 means that when the model predicted Real, it was correct 85.7% of the time. AI precision of 0.997 means that when the model predicted AI-generated, it was correct 99.7% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
DALLE3	7,500	7,467	33	0.996	0.995	1.000
SDXL	7,500	7,447	53	0.993	0.991	1.000
SD3	7,500	7,307	193	0.974	0.970	1.000
Midjourney	7,500	7,106	394	0.947	0.944	1.000
SD21	7,500	6,942	558	0.926	0.918	1.000

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
Real	7,500	0.015	0.000	4e-06	0.999991
AI-generated	37,500	0.964	1.000	5e-06	0.999995

MoLD Layer Weights

Layer	Average Gate Weight
block_11	0.312
block_10	0.230
block_8	0.118
block_9	0.072
block_4	0.062
block_1	0.055
block_6	0.039
block_7	0.038
block_0	0.030
block_5	0.018
block_2	0.015
block_3	0.014

Appendix A6: Finetuned All-Layer Contrastive Language-Image Pre-training (CLIP) MoLD

Field	Value
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Notebook	06_clip_mold_model_frozen_backbone_STANDARDIZED.ipynb
Final model name	final_clip_vit_l14_frozen_all_layer_mold
Architecture	Frozen CLIP ViT-L/14 vision encoder with trainable all-layer MoLD head

Final Parameters

Parameter	Value
Max epochs	15
Learning rate	1e-4
Weight decay	1e-4
Threshold	0.35
Batch size	16
Weighted sampler	True
Freeze backbone	True
Projection dim	512
Dropout	0.30
Layers used	All 24 CLIP vision transformer layers
Combination method	Learned softmax gate

Optimizer	AdamW
Loss	BCEWithLogitsLoss

Classification Report

Class/Metric	Precision	Recall	F1-score	Support
Real	0.907	0.998	0.950	7,500
AI-generated	1.000	0.979	0.989	37,500
Accuracy			0.983	45,000
Macro average	0.953	0.989	0.970	45,000
Weighted average			0.983	45,000

Summary Statistics

Metric	Value
Accuracy	0.983
Real precision	0.907
Real recall	0.998
Real F1	0.950
AI precision	1.000

AI recall	0.979
AI F1	0.989
Macro precision	0.953
Macro recall	0.989
Macro F1	0.970
Weighted F1	0.983
ROC AUC	0.999

Confusion Matrix and Intuitive Meaning

True label	Predicted Real	Predicted AI-generated
Real	7,486	14
AI-generated	770	36,730

In real terms, this model correctly classified 7,486 of 7,500 real images as Real and incorrectly flagged 14 real images as AI-generated. It correctly caught 36,730 of 37,500 AI-generated images and missed 770 AI-generated images by labeling them Real. Real precision of 0.907 means that when the model predicted Real, it was correct 90.7% of the time. AI precision of 1.000 means that when the model predicted AI-generated, it was correct 100.0% of the time.

Per-Generator AI Recall

AI Generator	Test Images	Correctly Predicted AI	Missed as Real	AI Recall	Mean AI Prob.	Median AI Prob.
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SDXL	7,500	7,485	15	0.998	0.997	1.000
DALLE3	7,500	7,470	30	0.996	0.995	1.000
SD3	7,500	7,469	31	0.996	0.995	1.000
SD21	7,500	7,378	122	0.984	0.978	1.000
Midjourney	7,500	6,928	572	0.924	0.916	1.000

Probability Summary

True Class	Images	Mean Predicted AI Probability	Median	Min	Max
Real	7,500	0.002	0.000	7.20144e-11	0.999164
AI-generated	37,500	0.976	1.000	7.29754e-09	1

MoLD Layer Weights

Layer	Average Gate Weight
layer_12	0.144
layer_13	0.083
layer_14	0.066
layer_22	0.063
layer_20	0.061
layer_10	0.059

layer_15	0.055
layer_9	0.054
layer_11	0.052
layer_8	0.050
layer_16	0.040
layer_21	0.039
layer_24	0.038
layer_23	0.034
layer_19	0.032
layer_17	0.032
layer_18	0.026
layer_7	0.024
layer_6	0.012
layer_2	0.011
layer_5	0.008
layer_3	0.007
layer_4	0.006

layer_1	0.005
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